

1. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 4 \end{bmatrix}$. Calculate each of the following expressions or explain why it is not defined.

a. AB

b. CB

2. Suppose that $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is defined by $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 \\ x_2 + x_3 \end{bmatrix}$.

a. Find the standard matrix A for T .

b. Is T is a linear transformation.

3. Let $\mathbf{x} \in \mathbb{R}^n$ and A be an $m \times n$ matrix. The function $f(\mathbf{x}) = A\mathbf{x}$ is a linear map from ___ to ___.

4. Find A^{-1} or show that it does not exist.

a. $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

b. $A = \begin{bmatrix} 2 & 2 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

5. Let $A = \begin{bmatrix} 1/2 & 1/4 \\ 0 & 1/2 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

a. Use A^{-1} to solve $A\mathbf{x} = \mathbf{b}$ for $\mathbf{x}$.

b. Use your answer from part (a) to express $\mathbf{b}$ as a linear combination of A .

c. Find $(A^{-1})^{-1}$ or show that it does not exist.

6. Let $\mathbf{v}_1 = (1, 2)$, $\mathbf{v}_2 = (0, 1)$, and $\mathbf{v}_3 = (1, 0)$ be three vectors in $\mathbb{R}^2$.

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- a. Show that $\mathbf{v}_1 = (1, 2)$ and $\mathbf{v}_2 = (0, 1)$ gives a basis for $\mathbb{R}^2$.
- b. Find another basis for $\mathbb{R}^2$.
- c. Is $\mathbf{v}_1 = (1, 2)$, $\mathbf{v}_2 = (0, 1)$, and $\mathbf{v}_3 = (1, 0)$ linearly independent? Explain.
- d. Is $\mathbf{v}_1 = (1, 2)$, $\mathbf{v}_2 = (0, 1)$, and $\mathbf{v}_3 = (1, 0)$ a basis for $\mathbb{R}^2$? Explain.
- e. Does $\mathbf{v}_1 = (1, 2)$, $\mathbf{v}_2 = (0, 1)$, and $\mathbf{v}_3 = (1, 0)$ span $\mathbb{R}^2$? Explain.
7. Let $\mathbf{v}_1 = (2, 4, 8)$, $\mathbf{v}_2 = (-2, -4, -8)$, $\mathbf{v}_3 = (1, 2, 4)$ be vectors in $\mathbb{R}^3$.
- a. Is $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ a subspace of $\mathbb{R}^3$?
- b. Is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ linearly independent?
- c. Does $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ span $\mathbb{R}^3$, i.e., does $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = \mathbb{R}^3$?
- d. Find the dimension of $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$.
- e. Find a basis for $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$.
8. Let $V = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n+1}\}$ be a set of $n + 1$ vectors in $\mathbb{R}^n$.
- a. Is V linearly independent or linearly dependent?
- b. Is V a basis for $\mathbb{R}^n$? Explain.
- c. Is $\text{span}V$ a subspace of $\mathbb{R}^n$. Explain.
- d. Is V necessarily a subspace of $\mathbb{R}^n$, i.e., is a set of $n + 1$ vectors in $\mathbb{R}^n$ always a subspace of $\mathbb{R}^n$? Explain.
9. **Bonus.** Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and let $\mathbf{v} \in \mathbb{R}^n$. Prove that $\mathbf{v} \in \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) \iff \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v})$.